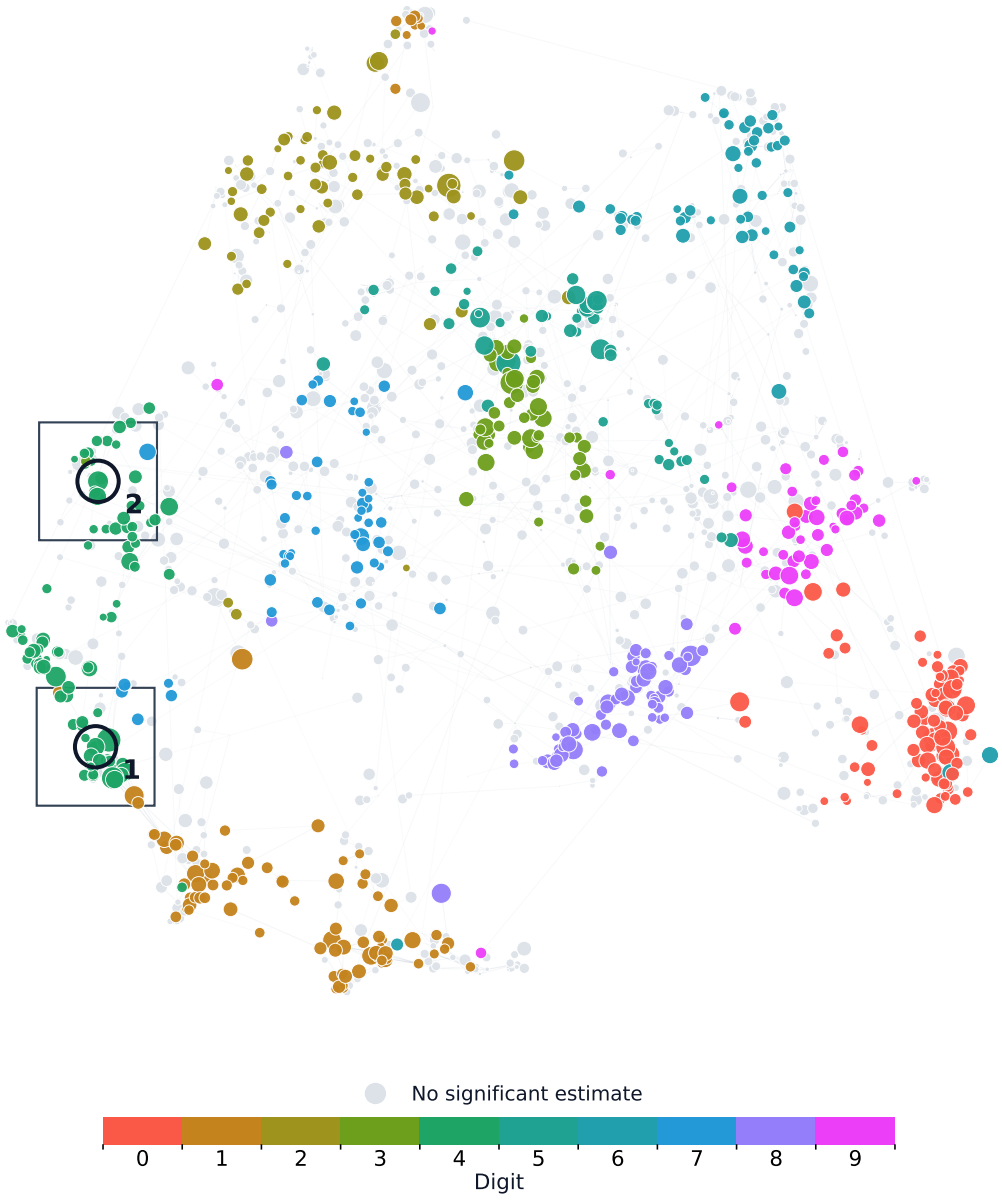
GLYPH / DISCOVERY BRIEF

Glyph + DINOv3: handwritten digits

680 significant associations · 1629 concepts mapped

24/545 evaluable hits reproduce in test; 135 discovery hits lack evaluable comparison estimates.

train 700 / val 150 / test 150
Reconstruction R^2: validation 0.868; test 0.864 · k=64
Test probe AUC: input vectors 0.996; full dictionary 1.000

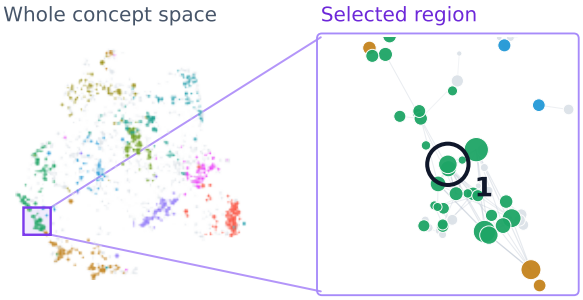


All training-alive concepts learned from public DINOv3 digit embeddings. Color: strongest significant outcome. Node area is proportional to activation prevalence, with a tiny visibility floor and a common density adjustment. Missing prevalences use a uniform small marker. Lines show saved correlations, with at most eight per concept.

Public UCI optical digits via sklearn.load_digits. Probe AUC shown is digit 0 versus rest; all ten outcome probes are saved. Associations are unadjusted. No automated interpretation pipeline or literature comparison was run. Style descriptions are unvalidated training-image observations.

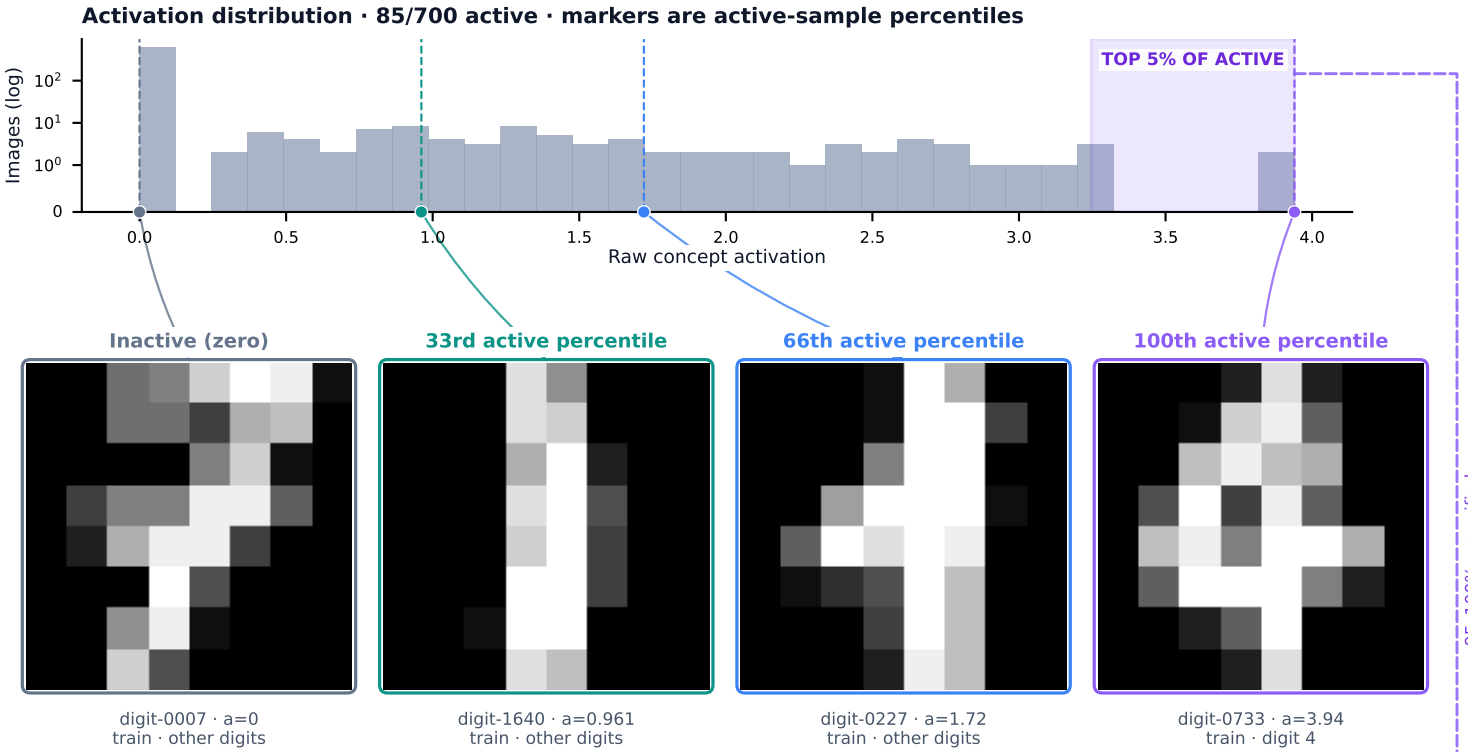
GLYPH / CONCEPT EVIDENCE

image_path concept 1978 → is_digit_4



train: OR 2.40 (95% CI 1.97–2.93); corrected p=8.9e-14
test: OR 2.41 (95% CI 1.34–4.32); corrected p=1

Compact, angular 4s with an upright central stroke among the strongest training examples. Visual description only; no subtype validation.

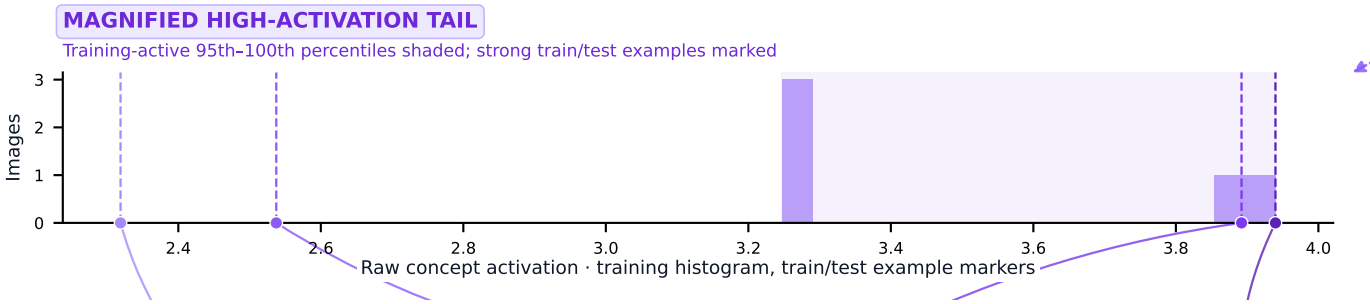


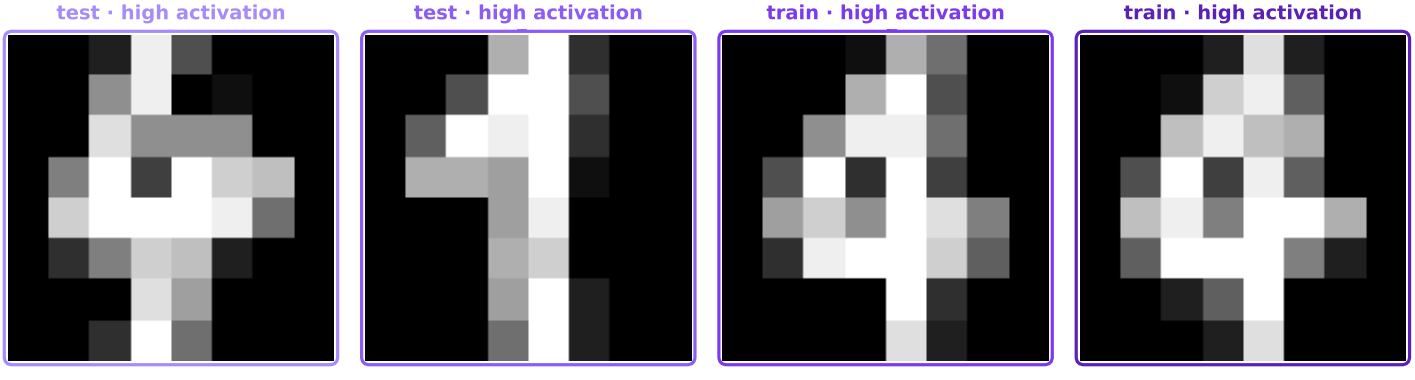
digit-0007 · a=0
train · other digits

digit-1640 · a=0.961
train · other digits

digit-0227 · a=1.72
train · other digits

digit-0733 · a=3.94
train · digit 4





digit-1526 · a=2.32
test · digit 4

digit-1298 · a=2.54
test · other digits

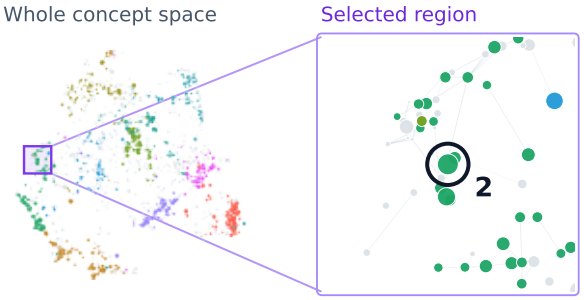
digit-0714 · a=3.89
train · digit 4

digit-0733 · a=3.94
train · digit 4

Scale images selected on training activations only. Zero is shown separately; other percentiles exclude zeros. Strong examples are selected by activation within train/test, without outcome-based selection. Images do not validate a semantic interpretation.

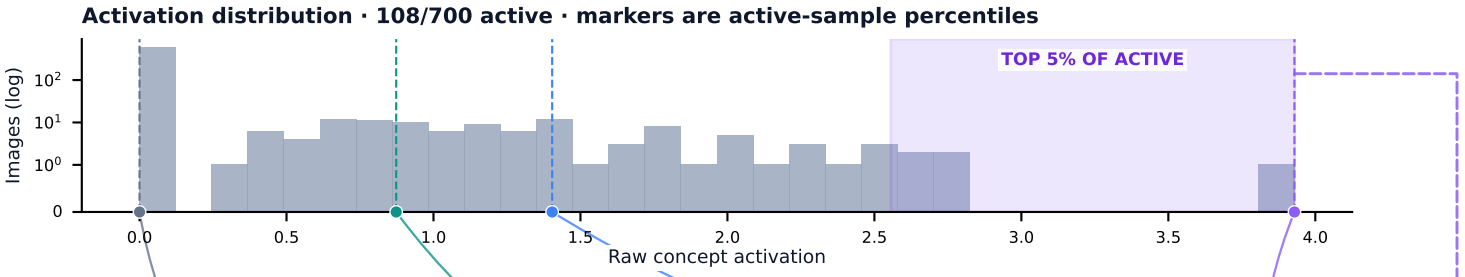
GLYPH / CONCEPT EVIDENCE

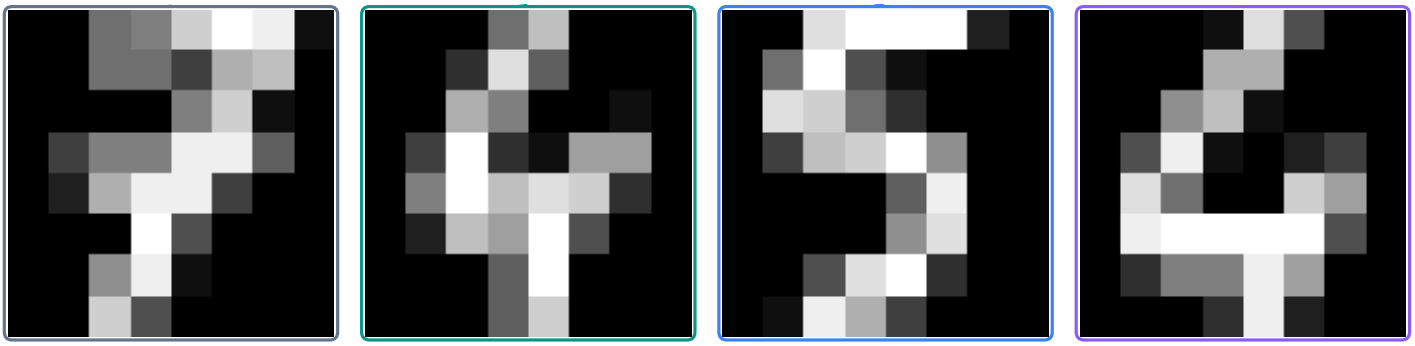
image_path concept 5058 → is_digit_4



train: OR 2.63 (95% CI 2.15–3.23); corrected p=1.9e-16
test: OR 3.18 (95% CI 1.98–5.12); corrected p=0.025

Broader, rounded/open-bowl 4s among the strongest training examples. Visual description only; no subtype validation.





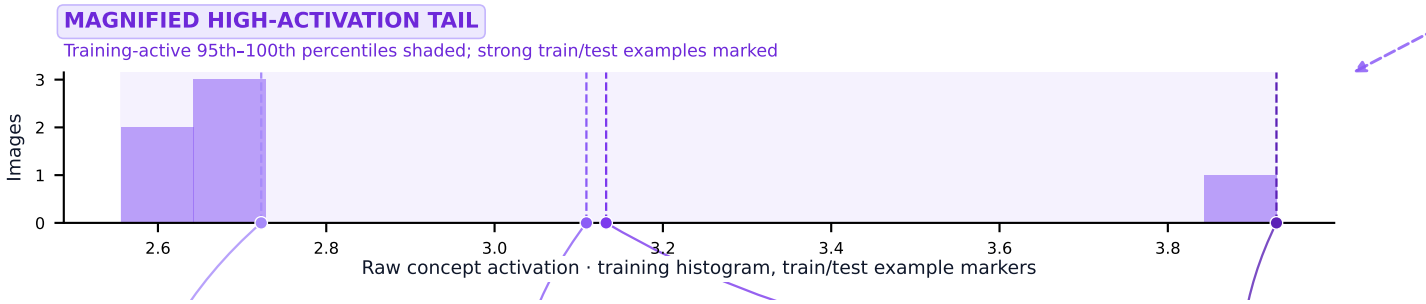
digit-0007 · a=0
train · other digits

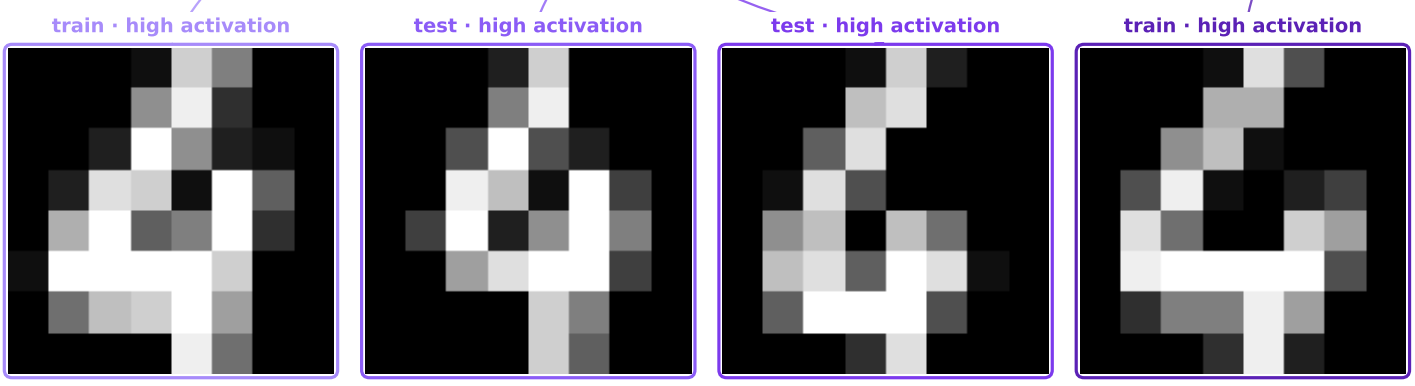
digit-0367 · a=0.873
train · digit 4

digit-1407 · a=1.4
train · other digits

digit-1221 · a=3.93
train · digit 4

95–100% magnified





digit-1778 · a=2.72
train · digit 4

digit-0100 · a=3.11
test · digit 4

digit-0876 · a=3.13
test · digit 4

digit-1221 · a=3.93
train · digit 4

Scale images selected on training activations only. Zero is shown separately; other percentiles exclude zeros. Strong examples are selected by activation within train/test, without outcome-based selection. Images do not validate a semantic interpretation.